

Elias Ramzi | AI Research Scientist

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End-to-end driving — planning, world models, and multimodal models

Experiences

valeo.ai (Paris, France) - full-time

2024-

AI research scientist

Deep learning research interests: end-to-end planning, world models, VLMs / LLMs.

- Developed world action models for autonomous driving, VaViM/VaVAM (technical report, CoRL Workshop, initial baseline for AlpaSim).
- Scaled training of generative models up to 1.9B parameters across 192 H100s using μ P and DeepSpeed ZeRO-2.
- Built Pictura, a GPU-accelerated multi-agent driving simulator rendering every agent's egocentric view, which unlocks perspective-view self-play at scale for driving policies (ECCV 2026 workshop paper).
- Co-supervising two PhD students: one on LLMs and VLMs (ICLR 2025, ACL Findings 2026), another on world models and RL.

Cnam & Coexya (Paris, France) - 3 year PhD

2021-2024

PhD student

PhD focused on computer vision and image retrieval:

- Designed novel ranking losses for image retrieval, combining theoretical analysis with large-scale experiments.
- Published four papers in major international machine learning conferences (NeurIPS, ECCV $\times$ 2, ICML), three of which as first author, and 2 papers in journals (TPAMI, TMLR).
- Transferred HAPPIER (ECCV) to trademark logo retrieval; the resulting models were deployed in production in Coexya's Acsepto product in 2024.
- Ran distributed training experiments across up to 16 GPUs on France's Jean Zay HPC cluster.

Cnam (Paris, France) - 5 month internship

2020

Research intern

Deep learning for 3D medical image segmentation. I used a learned confidence measure (ConfidNet) to combine segmentation results from different 2D U-Nets.

Sancare (Paris, France) - 6 month internship

2019

Data engineer and data scientist intern

Balto (Sydney, Australia) - 6 month internship

2018

Data engineer intern

Education

PhD in Computer Science, Cnam & Coexya (France)

2021-2024

Ranking losses and hierarchical learning for image retrieval. Awarded the "Prix de Thèse" by AFRIF ☑ (June 2025). 6 publications including 4 first-author papers.

Supervisor: Nicolas Thome ☑ (Sorbonne Université).

Advisors: Nicolas Audebert ☑ (IGN); Clément Rambour ☑ (Sorbonne Université).

Industrial advisor: Xavier Bitot ☑ (Coexya).

Engineering diploma, CentraleSupélec (France)

2016-2020

Courses in signal processing (sound, image, speech), machine learning, statistical models (estimators, Bayesian learning), data science (data mining, sparse data representation, NLP).

Technical skills

Open source projects

- Pictura simulator: 🤖 Pictura
- Overleaf MCP server: 🤖 WebLatexMCP
- Reproduce VaViM and VaVAM (CoRL'25): 🤖 VideoActionModel
- Global and local prompts for VLMs (ECCV'24): 🤖 GalLoP
- Out-of-distribution detection with HEAT (ICML'23): 🤖 HEAT.
- Ranking losses and hierarchical image retrieval (NeurIPS'21, ECCV'22, TPAMI'25): 🤖 ROADMAP, 🤖 HAPPIER, 🤖 SupRank & the hierarchical 🤖 google-landmark dataset.

Technologies

- My programming language: Python – PyTorch, Lightning, NumPy, Hydra, Pandas, Scikit-learn.
- My work environment : Linux/macOS + Git + VS Code & claude-code.
- Other tools: L^AT_EX, Shell script, SLURM, HyperQueue, Nextflow.

Contributions to open-source

- PyTorch metric learning: 🤖 pytorch-metric-learning.
- Implementation of Mixture of Experts: 🤖 mixture-of-experts.
- PyTorch-Lightning: 🤖 lightning.
- Universal Image Embeddings: 🤖 Universal-Image-Embeddings.

Miscellaneous

- French is my mother tongue and I speak fluently English (last tested at C1 level).
- I enjoy (trail) running, biking (🤖 ClaudeCyclingMCP) and back-country skiing.

Publications

Yuan Yin, **Elias Ramzi**, Marc Lafon, Valentin Charraut, Victor Bares, Yihong Xu, Éloi Zablocki, Alexandre Boulch, Thibault Buhet, Andrei Bursuc, *et al.* "Pictura: Perspective-View Self-Play at Scale for Driving." *arXiv preprint*, 2026. online: arxiv.org/abs/2607.26005

Yihong Xu, Éloi Zablocki, Yuan Yin, **Elias Ramzi**, Ellington Kirby, Alexandre Boulch, Matthieu Cord. "TOAD: Test-Time Trajectory Optimization for Autonomous Driving." *arXiv preprint*, 2026. online: arxiv.org/abs/2606.07170

Shashanka Venkataramanan, Valentin Pariza, Mohammadreza Salehi, Lukas Knobel, Spyros Gidaris, **Elias Ramzi**, Andrei Bursuc, Yuki M. Asano, "Franca: Nested Matryoshka Clustering for Scalable Visual Representation Learning." *CVPR*, 2026. online: arxiv.org/abs/2507.14137

Amaia Cardiel, Éloi Zablocki, **Elias Ramzi**, and Eric Gaussier. "DRIV-EX: Counterfactual Explanations for Driving LLMs." *ACL findings*, 2026. online: arxiv.org/abs/2603.00696

Florent Bartoccioni, **Elias Ramzi**, *et al.* "VaViM and VaVAM: Autonomous Driving through Video Generative Modeling" *Technical report & CoRL workshop*, 2025. online: arxiv.org/abs/2502.15672

Amaia Cardiel, Éloi Zablocki, **Elias Ramzi**, Oriane Siméoni, and Matthieu Cord. "LLM-wrapper: Black-Box Semantic Aware Adaptation of Vision-Language Models for Referring Expression Comprehension." *ICLR*, 2025. online: arxiv.org/abs/2409.11919

Elias Ramzi, Nicolas Audebert, Clément Rambour, André Araujo, Xavier Bitot and Nicolas Thome. "Optimization of Rank Losses for Image Retrieval." *TPAMI*, 2025. online: arxiv.org/abs/2309.08250

Marc Lafon, **Elias Ramzi**, Clément Rambour, Nicolas Audebert, Nicolas Thome. "GalLoP: Learning Global and Local Prompts for Vision-Language Models." *ECCV*, 2024. online: arxiv.org/abs/2407.01400

Yannis Karmim, **Elias Ramzi**, Raphaël Fournier S'niehotta, and Nicolas Thome. "ITEM: Improving Training and Evaluation of Message-Passing based GNNs for top-k recommendation." *TMLR*, 2024. online: arxiv.org/abs/2407.07912

Marc Lafon, **Elias Ramzi**, Clément Rambour, Nicolas Thome. "Hybrid Energy Based Model in the Feature Space for Out-of-Distribution Detection." *ICML*, 2023. online: arxiv.org/abs/2305.16966

Elias Ramzi, Nicolas Audebert, Nicolas Thome, Clément Rambour, and Xavier Bitot. "Hierarchical Average Precision Training for Pertinent Image Retrieval." *ECCV*, 2022. online: arxiv.org/abs/2207.04873

Elias Ramzi, Nicolas Thome, Clément Rambour, Nicolas Audebert, and Xavier Bitot. "Robust and Decomposable Average Precision for Image Retrieval." *NeurIPS*, 2021. online: arxiv.org/abs/2110.01445